Conducting Multiple Linear Regression in R

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Abstract

Multiple linear regression (MLR) is one of the most widely used statistical techniques in quantitative research. It models the relationship between a continuous outcome variable and two or more predictor variables. This handout complements the presentation by providing theoretical context, R code examples, interpretation guidance, diagnostics, and a short class exercise. The focus is on practical implementation in R using the lm() function.

Conducting Multiple Linear Regression in R

1 Introduction

Multiple linear regression (MLR) is one of the most widely used statistical techniques in quantitative research. It models the relationship between a continuous outcome variable and two or more predictor variables. Unlike simple linear regression, which includes only one predictor, multiple linear regression allows researchers to control for several factors at the same time. This makes it possible to examine how each predictor is related to the outcome while holding the other predictors constant (Field, 2018; James et al., 2021).

The general form of a multiple linear regression model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

where Y is the dependent variable, X_1 through X_k are independent variables, β_0 is the intercept, β_1 through β_k are regression coefficients, and ε is the error term.

This handout complements the presentation by providing theoretical context, R code examples, interpretation guidance, diagnostics, and a short class exercise. The focus is on practical implementation in R using the `lm()` function.

2 Data Preparation in R

Before running a regression model, the data should be loaded and checked for common problems such as missing values, incorrect variable types, and unusual observations. Good data preparation is important because regression results can be misleading if the input data is incomplete or incorrectly structured (Wickham et al., 2023).

2.1 Creating or Loading Data

For demonstration purposes, a small dataset can be created directly in R using `data.frame()`. In applied research, data is often imported from external files, for example with `read.csv()` for CSV files or `readxl::read_excel()` for Excel files.

```
data <- data.frame(
  Salary     = c(2000, 2500, 3000, 3500, 4000),
  Age        = c(22, 25, 30, 35, 40),
  Education  = c(12, 14, 16, 16, 18),
  Experience = c(1, 3, 5, 7, 10)
)
```

```
data
```

```
  Salary Age Education Experience
1   2000  22        12          1
2   2500  25        14          3
3   3000  30        16          5
4   3500  35        16          7
5   4000  40        18         10
```

For example, a researcher studying income determinants could collect survey data containing salary, years of schooling, age, and work experience from employees. In practice, such a dataset could be imported like this:

```
salary_data <- read.csv("salary_data.csv")
```

2.2 Inspecting the Data

After loading the data, several R functions can be used to inspect the dataset before estimating the model.

```
head(data)
```

```
  Salary Age Education Experience
1   2000  22        12          1
2   2500  25        14          3
3   3000  30        16          5
4   3500  35        16          7
5   4000  40        18         10
```

```
summary(data)
```

```
     Salary          Age         Education       Experience
 Min.   :2000   Min.   :22.0   Min.   :12.0   Min.   : 1.0
 1st Qu.:2500   1st Qu.:25.0   1st Qu.:14.0   1st Qu.: 3.0
```

```
 Median :3000   Median :30.0   Median :16.0   Median : 5.0  
 Mean   :3000   Mean   :30.4   Mean   :15.2   Mean   : 5.2  
 3rd Qu.:3500   3rd Qu.:35.0   3rd Qu.:16.0   3rd Qu.: 7.0  
 Max.   :4000   Max.   :40.0   Max.   :18.0   Max.   :10.0  
```

```r
str(data)
```

```
'data.frame':   5 obs. of  4 variables:
 $ Salary    : num  2000 2500 3000 3500 4000
 $ Age       : num  22 25 30 35 40
 $ Education : num  12 14 16 16 18
 $ Experience: num  1 3 5 7 10
```

```r
is.na(data)
```

```
     Salary   Age Education Experience
[1,]  FALSE FALSE     FALSE      FALSE
[2,]  FALSE FALSE     FALSE      FALSE
[3,]  FALSE FALSE     FALSE      FALSE
[4,]  FALSE FALSE     FALSE      FALSE
[5,]  FALSE FALSE     FALSE      FALSE
```

The function `head()` shows the first rows of the data, `summary()` provides descriptive statistics, `str()` displays the variable types, and `is.na()` checks for missing values. In the salary example, the dataset contains five observations with salary as the outcome variable and age, education, and experience as predictors.

3 Model Specification and Estimation

In R, regression models are specified using a formula interface. The dependent variable is placed on the left side of the tilde symbol, and the predictors are placed on the right side.

The model formula used in this handout is:

```
Salary ~ Age + Education + Experience
```

The model is estimated with the `lm()` function, which fits a linear model using ordinary least squares.

```
model <- lm(Salary ~ Age + Education + Experience, data = data)
model
```

```
Call:
lm(formula = Salary ~ Age + Education + Experience, data = data)

Coefficients:
(Intercept)          Age    Education   Experience
   -655.738       81.967       73.770        8.197
```

The object `model` contains the estimated coefficients, residuals, fitted values, and other diagnostic information.

```
fitted(model)
```

```
       1        2        3        4        5
2040.984 2450.820 3024.590 3450.820 4032.787
```

```
residuals(model)
```

```
        1         2         3         4         5
-40.98361  49.18033 -24.59016  49.18033 -32.78689
```

4 Interpreting the Output

The main regression output can be obtained with `summary(model)`.

```
summary(model)
```

```
Call:
lm(formula = Salary ~ Age + Education + Experience, data = data)

Residuals:
     1      2      3      4      5
-40.98  49.18 -24.59  49.18 -32.79

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -655.738   2029.578  -0.323    0.801
Age           81.967     74.676   1.098    0.470
Education     73.770     82.783   0.891    0.537
Experience     8.197    180.886   0.045    0.971

Residual standard error: 90.54 on 1 degrees of freedom
Multiple R-squared:  0.9967,    Adjusted R-squared:  0.9869
F-statistic: 101.3 on 3 and 1 DF,  p-value: 0.07287
```

The most important parts of the output are the coefficient estimates, standard errors, t values, p values, R-squared, adjusted R-squared, and the F-statistic.

For example, a positive coefficient for `Experience` would mean that higher work experience is associated with a higher predicted salary, assuming age and education remain constant. This interpretation follows the logic of multiple regression: each coefficient is interpreted while controlling for the other predictors in the model (Field, 2018).

5 Model Diagnostics

After estimating a regression model, it is important to check whether the assumptions of ordinary least squares are reasonably met. Diagnostic plots help assess linearity, normality of residuals, constant variance, and influential observations.

```r
par(mfrow = c(2, 2))
plot(model)
```

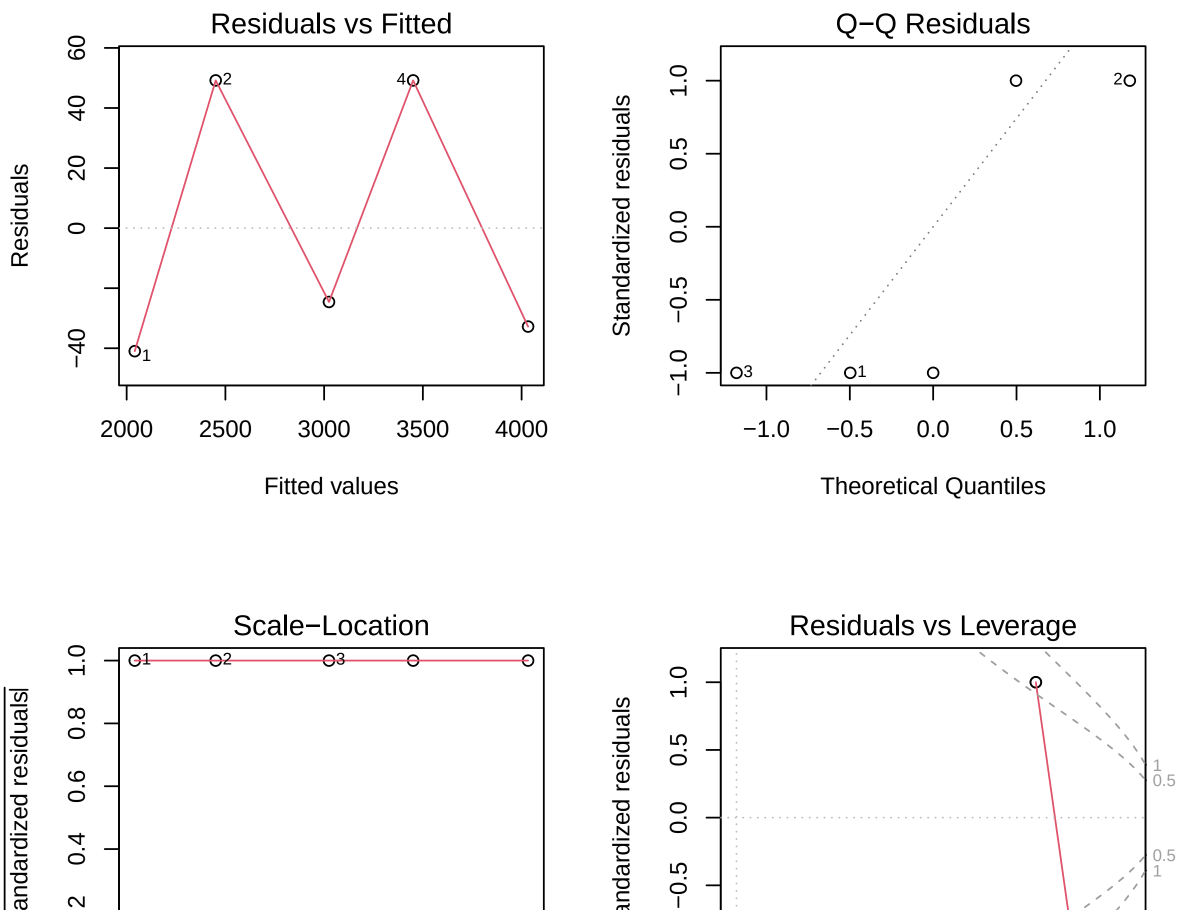
Table 1

Summary of regression output components

Component	Interpretation
Estimate	Expected change in the dependent variable for a one-unit increase in the predictor, holding other predictors constant.
Standard Error	Indicates the precision of the coefficient estimate. Smaller values suggest more precise estimates.
t value	Ratio of the estimate to its standard error. Larger absolute values indicate stronger evidence against the null hypothesis.
p value	Tests whether the coefficient is statistically different from zero. A common threshold is $p < .05$.
R-squared	Proportion of variance in the dependent variable explained by the model.
Adjusted R-squared	R-squared adjusted for the number of predictors. Useful when comparing models.
F-statistic	Tests whether the model explains a significant amount of variance overall.

Figure 1

Diagnostic plots for the multiple linear regression model.



```
par(mfrow = c(1, 1))
```

The four standard diagnostic plots can be interpreted as follows:

- **Residuals vs. Fitted:** Checks whether the relationship is approximately linear. Ideally, residuals should be randomly scattered around zero.
- **Normal Q-Q Plot:** Checks whether residuals are approximately normally distributed. Points should follow the diagonal line.
- **Scale-Location Plot:** Checks whether residual variance is constant. A funnel shape may indicate heteroscedasticity.
- **Residuals vs. Leverage:** Helps identify influential observations that may strongly affect the regression results.

If the diagnostic plots show strong violations, the researcher may need to transform variables, remove or investigate influential observations, or consider alternative models.

6 Making Predictions

Once a model has been fitted, it can be used to generate predictions for new observations. The new data frame must contain the same predictor names used in the model.

```
new_employee <- data.frame(
  Age        = 33,
  Education  = 16,
  Experience = 6
)

predict(model, newdata = new_employee)
```

```
       1
3278.689
```

A prediction interval can also be requested. This gives a range in which a future individual observation is expected to fall.

```
predict(
  model,
  newdata = new_employee,
```

```
  interval = "prediction",
  level = 0.95
)
```

```
       fit      lwr      upr
1 3278.689 1641.855 4915.522
```

For example, an HR manager could use the fitted model to estimate the expected salary for a new employee with a specific age, education level, and amount of work experience. The prediction interval communicates uncertainty around the point estimate.

7 Class Exercise

Complete the following exercise using R. The goal is to practice specifying and interpreting a multiple linear regression model with your own variables.

7.1 Task

Create a data frame with at least eight observations and at least two predictor variables. For example, you could model house price using size and number of rooms, or student GPA using study hours and attendance rate.

Then complete the following steps:

- Fit a multiple linear regression model.
- Use `summary()` to examine the output.
- Identify which predictors are statistically significant at the 5% level.
- Report the adjusted R-squared and explain what it says about model fit.
- Interpret at least one coefficient in plain language.
- Run diagnostic plots and describe what they suggest about the regression assumptions.
- Use `predict()` to estimate the outcome for a new observation.

7.2 Starter Template

```
my_data <- data.frame(
  y  = c(...),  # outcome variable
  x1 = c(...),  # first predictor
  x2 = c(...)   # second predictor
)

head(my_data)
```

```
summary(my_data)

my_model <- lm(y ~ x1 + x2, data = my_data)

summary(my_model)

par(mfrow = c(2, 2))
plot(my_model)

new_obs <- data.frame(
  x1 = ...,
  x2 = ...
)

predict(my_model, newdata = new_obs)
```

8 Comprehensive End-to-End Example

This section summarizes the complete workflow from data creation to prediction. It mirrors the practical steps covered in the presentation and can be used as a reference template.

8.1 Scenario

A company collects data on five employees and wants to model monthly salary in euros as a function of age, years of education, and years of work experience.

8.2 Step 1: Load or Create Data

```
data <- data.frame(
  Salary     = c(2000, 2500, 3000, 3500, 4000),
  Age        = c(22, 25, 30, 35, 40),
  Education  = c(12, 14, 16, 16, 18),
  Experience = c(1, 3, 5, 7, 10)
)
```

8.3 Step 2: Inspect the Data

```
head(data)
summary(data)
str(data)
```

8.4 Step 3: Specify and Estimate the Model

```r
model <- lm(Salary ~ Age + Education + Experience, data = data)
```

8.5 Step 4: Extract Results

```r
summary(model)
```

The key output includes the coefficients, p values, R-squared, adjusted R-squared, and F-statistic.

8.6 Step 5: Run Diagnostics

```r
par(mfrow = c(2, 2))
plot(model)
```

The diagnostic plots help assess linearity, normality, homoscedasticity, and influential observations.

8.7 Step 6: Predict for a New Employee

```r
new_employee <- data.frame(
  Age        = 33,
  Education  = 16,
  Experience = 6
)

predict(model, newdata = new_employee)

predict(
  model,
  newdata = new_employee,
  interval = "prediction",
  level = 0.95
)
```

This six-step workflow — loading, inspecting, specifying, estimating, diagnosing, and predicting — constitutes the core practical competency in applied regression analysis using R.

9 Presentation and Handout Links

GitHub link to the handout: INSERT_HANDOUT_GITHUB_LINK

GitHub link to the presentation: INSERT_PRESENTATION_GITHUB_LINK

10 References

Field, A. (2018). *Discovering statistics using IBM SPSS statistics* (5th ed.). SAGE Publications.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in r* (2nd ed.). Springer. https://doi.org/10.1007/978-1-0716-1418-1

Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science: Import, tidy, transform, visualize, and model data* (2nd ed.). O'Reilly Media.

Appendix

Affidavit

I hereby affirm that this submitted paper was authored unaided and solely by me. Additionally, no other sources than those listed in the reference list were used. Parts of this paper, including tables and figures, that have been taken either verbatim or analogously from other works have in each case been properly cited with regard to their origin and authorship.

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